OUTDOOR LED DURABILITY CHECKLIST

UAE/GCC outdoor specification checklist — heat, dust, IP, service | BRIGHTLUXX

IP RATING

Requirement	Notes
IP65 front, IP54 rear (minimum)	Protects against Gulf sand/dust + rain
IP65 full if exposed to direct rain	Pool areas, unprotected facades
Conformal coating on PCBs	Salt air corrosion protection (coastal)

THERMAL MANAGEMENT

Requirement	Notes
Operating range: -10C to 60C	UAE summer ambient reaches 50C+
Smart fan speed control	Temperature-based automatic adjustment
Thermal protection auto-dimming	Reduces brightness to prevent damage
Rear ventilation gap: min 200mm	Essential for heat dissipation

BRIGHTNESS & VISIBILITY

Requirement	Notes
Minimum 5,500 nits for shade	North-facing, canopy, covered areas
7,000-10,000 nits for direct sun	South and west-facing installations
Anti-glare coating / matte surface	Reduces reflection, improves readability
Auto-brightness sensor	Adjusts to ambient light automatically

STRUCTURAL & ELECTRICAL

Requirement	Notes
Wind load calculation (local code)	UAE can have sandstorms, strong gusts
Stainless steel or galvanized structure	Prevents corrosion in Gulf humidity
Surge protection (SPD)	Essential — UAE has power fluctuations
Dedicated electrical circuit + 20%	Separate DB, MCBs, RCDs

SERVICE ACCESS

Requirement	Notes
Front-service cabinets preferred	Allows maintenance from scaffold/cherry picker
Tool-free module removal	Reduces service time and risk
Spare parts: 5% modules, 2x PSU	Same-batch modules for colour match
Remote monitoring capability	Alert on failures, temperature, status